

Full Adder Design in Verilog HDL

Overview

This project implements a **1-bit Full Adder** using **Verilog HDL** and verifies its functionality through simulation in **Xilinx Vivado**.

Features

- Verilog implementation of a 1-bit Full Adder
- Testbench covering all 8 input combinations
- Behavioral simulation with waveform verification
- Demonstrates correct generation of **Sum** and **Carry** outputs

Files

- **full_adder02.v** – Full Adder design module
- **stimulation.v** – Testbench for simulation
- Simulation waveforms and project files

Truth Table

A	B	Cin	Sum	Carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Tools Used

- Verilog HDL
- Xilinx Vivado
- Behavioral Simulation

Learning Outcome

This project demonstrates combinational logic design, Verilog coding practices, testbench creation, and waveform analysis for digital circuit verification.